

Using the vector product

Key Points

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$ is $|\mathbf{a} \times \mathbf{b}|$ and the triangle is $\frac{1}{2}|\mathbf{a} \times \mathbf{b}|$

The equation of a line going through $\mathbf{a}$ and $\mathbf{b}$ can be written as

$$(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$$

As parallel vectors have a zero vector product

Basic Skills

Q1. Find the vector product of

a) $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 4 \\ 4 \\ 1 \end{pmatrix}$

Q2. Find the magnitude of the vector $\begin{pmatrix} 5 \\ 3 \\ -8 \end{pmatrix}$

Q3. Find the two unit normals between the vectors $\begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 7 \end{pmatrix}$

Competency Skills

Q4. Explain why parallel vectors have a zero vector product

Q5. State the value of $\mathbf{i} \times \mathbf{j}, \mathbf{j} \times \mathbf{k}, \mathbf{k} \times \mathbf{i}$

Q6. Find the area of the parallelogram formed by the vectors $\mathbf{a} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$

Q7. Find the area of the triangle formed by the coordinates (5,3,2), (6,-2,10) and (5,-2,1)

Q8. Given the line is $\mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ 2 \end{pmatrix}$ write the equation in vector product form

Advanced Skills

Q9. Explain what the vector product represents geometrically

Q10. Draw a diagram to show why the area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$ is $|\mathbf{a} \times \mathbf{b}|$

Q11. Prove algebraically that $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$

Q12. For the coordinates A=(2,3,5), B=(3,-1,-5), C=(k,5,2) and D=(0,k,12) we know that AB is parallel to CD. Find the shortest distance between the lines AB and CD

Extension

Q13. Deduce a formula for the volume of a parallelepiped formed by vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$

Hint: Use the volume is the base multiplied by the perpendicular height